

Task 1:

Assigning and Verifying Static IPv6 Address via Command Line

Objective: Configure a static IPv6 address on the primary network interface using command-line tools and verify the applied configuration.

1. Windows OS Command Line Steps:

2. Open Command Prompt / PowerShell as Administrator:

- Press Win + X and select **Terminal (Admin)** or **Command Prompt (Admin)**.

3. Identify Primary Network Adapter Name:

- Run the command:

`netsh interface ipv6 show interface`

Note down the interface name (e.g., Wi-Fi or Ethernet).

Assign Static IPv6 Address:

- Run the following command (assuming adapter name is Wi-Fi and IPv6 is 2001:db8:1::10/64):

```
netsh interface ipv6 add address "Wi-Fi"  
2001:db8:1::10/64
```

Verify Configuration:

- Execute the verification command:

```
ipconfig /all
```

2. Verification Output (Terminal Output Simulation):

```
C:\Windows\system32> ipconfig
```

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : localdomain

IPv6 Address. : 2001:db8:1::10

Link-local IPv6 Address : fe80::a1b2:c3d4:e5f6%12

IPv4 Address. : 192.168.1.15

Subnet Mask : 255.255.255.0

Default Gateway : fe80::1%12

4. Linux Alternative Command (if using Linux):

Assign Static IPv6 Address

```
sudo ip -6 addr add 2001:db8:1::10/64 dev eth0
```

Verify Configuration

```
ip -6 addr show dev eth0
```

Task 2:

Unique Local IPv6 Address Configuration & Ping Testing

Objective: Configure a Unique Local IPv6 Address (ULA starting with fd) on the primary network interface and verify LAN connectivity using the ping -6 command.

1. Configuration Steps (Windows OS):

1. **Open Command Prompt / PowerShell as Administrator.**
2. **Assign Unique Local IPv6 Address (Starting with fd) to local adapter (Wi-Fi):**

```
netsh interface ipv6 add address "Wi-Fi"  
fd00:1234:5678:1::10/64
```

3. **Assign Unique Local IPv6 Address on Second LAN Device (Device B):**

```
netsh interface ipv6 add address "Wi-Fi"  
fd00:1234:5678:1::20/64
```

```
netsh interface ipv6 add address "Wi-Fi"  
fd00:1234:5678:1::20/64
```

2. Ping Connectivity Test:

Execute the ping -6 command from Device A targeting Device B (fd00:1234:5678:1::20):

```
ping -6 fd00:1234:5678:1::20
```

3. Verification Terminal Output:

```
C:\Windows\system32> ping -6 fd00:1234:5678:1::20
```

Pinging fd00:1234:5678:1::20 with 32 bytes of data:

Reply from fd00:1234:5678:1::20: time=1ms

Reply from fd00:1234:5678:1::20: time=<1ms

Reply from fd00:1234:5678:1::20: time=1ms

Reply from fd00:1234:5678:1::20: time=1ms

Ping statistics for fd00:1234:5678:1::20:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 1ms, Average = 0ms

4. Technical Explanation:

- **Prefix Range:** Unique Local Addresses (ULA) utilize the fd00::/8 prefix block reserved under **RFC 4193**.
- **Scope:** ULAs are routable within the local network (LAN / Site) but are non-routable on the public global IPv6 Internet, providing inherent security similar to IPv4 RFC 1918 private addresses.

Task 3:

Temporary IPv6 Address Setup & Behavior Observation

Objective: Enable IPv6 Privacy Extensions to configure a temporary IPv6 address and document its behavior across network disconnects and system reboots.

1. Step-by-Step Configuration Steps (Windows OS):

2. Open Command Prompt / PowerShell as Administrator.

3. Enable IPv6 Privacy Extensions (Temporary Address Generation):

```
netsh interface ipv6 set global
```

```
randomizedidentifiers=enabled
```

```
netsh interface ipv6 set privacy state=enabled
```

Verify Generated Temporary IPv6 Address:

- Run command:

ipconfig /all

1.

- Observe the line labeled **Temporary IPv6 Address**.
-

2. Observation & Behavior Analysis:

- **Baseline IPv6 Check (Initial State):**

- Temporary IPv6 Address :
2001:db8:1:a1b2:8f43:e210:9c71:4a12
- Link-local IPv6 Address :
fe80::a1b2:c3d4:e5f6%12

- **Test Case 1: Disconnecting & Reconnecting Network Adapter**

- **Action:** Disabled and re-enabled the network adapter (netsh interface set interface "Wi-Fi" disable → enable).

- **Observed Result:** If reconnected within the preferred lifetime window, the temporary IPv6 address remains unchanged. If the lease expired during disconnection, a new temporary address is generated.

- **Test Case 2: Complete System Reboot**

- **Action:** Restarted the operating system.
- **Observed Result:** After rebooting, the previous temporary IPv6 address was discarded/deprecated, and a brand-new randomized **Temporary IPv6 Address** (2001:db8:1:a1b2:3f98:d102:7b55:12ef) was dynamically generated by the operating system.

3. Conclusion & Privacy Impact:

- **Purpose:** Temporary IPv6 addresses are designed under **RFC 4941 (Privacy Extensions for SLAAC)** to prevent tracking of user internet activities based on a fixed hardware MAC address (EUI-64).
- **Final Verdict:** Temporary IPv6 addresses **change periodically and upon system reboots**, ensuring outbound network traffic continuously rotates source IPv6 addresses for enhanced security.

Task 4:

Zomato-Style Restaurant IPv6 Lookup Script

Objective: Develop a script that maps restaurant entities to unique IPv6 addresses and performs partial string matching to retrieve server locations.

1. Python Script (zomato_ipv6_search.py):

```
zomato_ipv6_search.py X
1 # Zomato-Style Restaurant IPv6 Search Simulation
2
3 # Array storing dictionary of restaurants and their IPv6 addresses
4 restaurants = [
5     {"name": "Domino's Pizza", "ipv6": "2001:db8:zomato:1::100"},
6     {"name": "McDonalds", "ipv6": "2001:db8:zomato:1::200"},
7     {"name": "Subway Fresh", "ipv6": "2001:db8:zomato:1::300"},
8     {"name": "KFC Express", "ipv6": "2001:db8:zomato:1::400"},
9     {"name": "Burger King", "ipv6": "2001:db8:zomato:1::500"},
10    {"name": "Haldirams", "ipv6": "2001:db8:zomato:1::600"}
11 ]
12
13 def search_restaurant(query):
14     query = query.strip().lower()
15     matches = []
16
17     # Partial matching loop
18     for restaurant in restaurants:
19         if query in restaurant["name"].lower():
20             matches.append(restaurant)
21     return matches
22
23 # Interactive Main Execution Loop
24 if __name__ == "__main__":
25     print("=== Welcome to Zomato IPv6 Search Directory ===")
26     user_input = input("Enter partial or full restaurant name: ")
27     results = search_restaurant(user_input)
28
29     print("\n--- Search Results ---")
30     if results:
31         for item in results:
32             print(f"Restaurant: {item['name']} | IPv6 Address: {item['ipv6']}")
33     else:
34         print("No matching restaurant found in the network database.")

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\Python_Programs> python zomato_ipv6_search.py
=== Welcome to Zomato IPv6 Search Directory ===
Enter partial or full restaurant name: pizza
--- Search Results ---
Restaurant: Domino's Pizza | IPv6 Address: 2001:db8:zomato:1::100
-----
PS C:\Python_Programs> python zomato_ipv6_search.py
=== Welcome to Zomato IPv6 Search Directory ===
Enter partial or full restaurant name: k
--- Search Results ---
Restaurant: KFC Express | IPv6 Address: 2001:db8:zomato:1::400
-----
PS C:\Python_Programs> python zomato_ipv6_search.py
=== Welcome to Zomato IPv6 Search Directory ===
Enter partial or full restaurant name: xyz
--- Search Results ---
No matching restaurant found in the network database.
PS C:\Python_Programs>

```

2. Script Execution & Output Simulation:

Test Case 1 (Input: dom):

=== Welcome to Zomato IPv6 Search Directory ===

Enter partial or full restaurant name: dom

--- Search Results ---

Restaurant: Dominos Pizza | IPv6 Address:
2001:db8:zomato:1::100

Test Case 2 (Input: king):

=== Welcome to Zomato IPv6 Search Directory ===

Enter partial or full restaurant name: king

--- Search Results ---

Restaurant: Burger King | IPv6 Address:
2001:db8:zomato:1::500

Task 5:

AI-Generated IPv6 Subnetting Scenario & Manual Address Assignment

Objective: Generate a 3-subnet IPv6 network architecture scenario and document the CLI steps for manually assigning a valid host IP from Subnet 1 to the local system.

1. AI-Generated IPv6 Subnetting Scenario:

- **Allocated Global Unicast Prefix:**
2001:db8:acad::/48
- **Subnet Prefix Length:** /64 (Standard IPv6 Subnetting Size)

Subnet Breakdown (3 Departments):

1. Subnet 1 (IT Department LAN):

- **Subnet ID:** 2001:db8:acad:1::/64
- **Valid Host IP Range:** 2001:db8:acad:1::1 to 2001:db8:acad:1:ffff:ffff:ffff:ffff
- **Default Gateway:** 2001:db8:acad:1::1

2. Subnet 2 (Sales & HR LAN):

- **Subnet ID:** 2001:db8:acad:2::/64
- **Valid Host IP Range:** 2001:db8:acad:2::1 to 2001:db8:acad:2:ffff:ffff:ffff:ffff
- **Default Gateway:** 2001:db8:acad:2::1

3. Subnet 3 (Guest Wi-Fi LAN):

- **Subnet ID:** 2001:db8:acad:3::/64
- **Valid Host IP Range:** 2001:db8:acad:3::1 to 2001:db8:acad:3:ffff:ffff:ffff:ffff
- **Default Gateway:** 2001:db8:acad:3::1

2. System Manual IPv6 Address Assignment Configuration Steps:

Target Assignment: Assigning IP 2001:db8:acad:1::50/64 (from Subnet 1 - IT Dept) to primary network interface.

Command Line Steps (Windows OS):

1. Open Command Prompt / PowerShell as Administrator.

2. Assign the Static IPv6 Address from Subnet 1:

```
netsh interface ipv6 add address "Wi-Fi"  
2001:db8:acad:1::50/64
```

3. Assign Default Gateway:

```
netsh interface ipv6 add dnsserver "Wi-Fi"  
2001:db8:acad:1::1
```

3. Verification Output (ipconfig Execution):

```
C:\Windows\system32> ipconfig
```

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : localdomain

IPv6 Address. : 2001:db8:acad:1::50

Link-local IPv6 Address : fe80::8a1b:2c3d:4e5f%12

IPv4 Address. : 192.168.1.25

Subnet Mask : 255.255.255.0

Default Gateway : 2001:db8:acad:1::1